King Fahad Central Hospital

Document ID: SU-011-PROT-01 | Parent Policy: SU-011 — Acute Stroke Immediate Management Pathway (0–24 Hour Clinical Pathway) | Version: 1.0 | Controlled Copy

Certification Alignment Note: Advanced imaging (including CT perfusion) is used when available and clinically indicated; the protocol must not imply a capability that is not operational at KFCH. Vessel imaging for LVO should not be delayed by unnecessary steps.

Acute Stroke CT Imaging Protocol

Non-Contrast CT, CT Angiography & Advanced Imaging (CT Perfusion/MRI when available and indicated) — AHA/ASA Guideline-Aligned

1. Purpose & Imaging Time Targets

Rapid, protocol-driven neuroimaging is central to acute stroke triage: it excludes hemorrhage, estimates infarct burden, identifies large vessel occlusion (LVO), and — in the extended time window — characterizes the ischemic core/penumbra to guide reperfusion therapy decisions.

Metric	Target
Door-to-CT initiation (non-contrast)	≤ 25 minutes from ED arrival
Door-to-CT interpretation	≤ 45 minutes from ED arrival
CT table notification	Stroke alert/code stroke activation triggers direct-to-scanner notification, bypassing routine queue

Targets reflect commonly cited AHA/ASA Target: Stroke initiative benchmarks — confirm current benchmark values against the active guideline/registry version.

2. Activation Workflow

- ☐ Stroke alert/code stroke activated by ED per LKW time and clinical screen — CT/CT technologist notified simultaneously with stroke team
- ☐ Patient transported directly to CT table on arrival, bypassing routine registration/imaging queue when protocol allows
- ☐ Stroke order set imaging orders (non-contrast CT ± CTA ± CTP) placed as a single bundled protocol to avoid sequential ordering delays
- ☐ Weight, renal function (if available), and contrast allergy history communicated to CT technologist before contrast administration
- ☐ IV access confirmed adequate for power injection (contrast studies) before patient reaches scanner

3. Non-Contrast CT Head

3.1 Purpose

- ☐ Exclude intracranial hemorrhage (absolute requirement before thrombolysis)
- ☐ Assess early ischemic changes and estimate infarct core burden via ASPECTS scoring
- ☐ Identify alternative diagnoses (mass, prior infarct, other structural lesion)

3.2 Acquisition (parameters set by Radiology/CT physics per scanner)

- ☐ Thin-section axial acquisition from skull base through vertex, reconstructed to standard soft-tissue and bone algorithm
- ☐ No IV contrast for this initial sequence
- ☐ Reconstruction slice thickness per departmental stroke protocol (typically thin sections for ASPECTS scoring plus standard thicker sections for routine read)

3.3 Interpretation — ASPECTS

Alberta Stroke Program Early CT Score: 10-point scoring system evaluating MCA territory; 1 point subtracted for each of 10 defined regions showing early ischemic change (score of 10 = no early changes; lower scores indicate larger territory involved).

- ☐ ASPECTS documented in the read and communicated verbally to the stroke team for time-critical decisions
- ☐ ASPECTS ≥6 is a standard eligibility criterion referenced in early-window thrombectomy selection (see institutional Mechanical Thrombectomy Selection Protocol)
- ☐ Hemorrhage, mass effect, or established large completed infarct documented and communicated immediately — may preclude thrombolysis/thrombectomy

4. CT Angiography (Head & Neck)

4.1 Purpose

- ☐ Identify and localize large vessel occlusion (intracranial ICA, MCA-M1/M2, basilar, vertebral)
- ☐ Assess collateral circulation status
- ☐ Evaluate cervical vasculature for tandem occlusion, dissection, or significant carotid stenosis relevant to intervention planning

4.2 Acquisition

- ☐ Coverage from aortic arch (or proximal great vessels per departmental protocol) through vertex
- ☐ IV contrast bolus via power injector through adequate-gauge peripheral IV, timed to arterial phase (bolus tracking or fixed-delay technique per departmental protocol)
- ☐ Single-phase CTA is standard; multiphase CTA (arterial, peak venous, late venous) may be used per departmental protocol to further characterize collateral status
- ☐ Thin-section reconstruction with multiplanar and maximum-intensity-projection (MIP) reformats for vessel evaluation

4.3 Contrast Safety Screening

- ☐ Renal function (creatinine/eGFR) reviewed per departmental contrast policy — do not delay emergent CTA in a thrombectomy candidate solely pending renal function unless departmental policy specifies otherwise; follow institutional emergent-contrast pathway
- ☐ Contrast allergy history reviewed; premedication per departmental protocol if history present and time allows, or per emergent allergy pathway
- ☐ Metformin use noted per departmental post-contrast renal monitoring policy
- ☐ IV hydration per departmental protocol if renal impairment identified and time permits

5. CT Venography (CTV) for Suspected Cerebral Venous Thrombosis (CVT/CVST)

- ☐ CTV is performed when cerebral venous thrombosis is clinically suspected or when non-contrast CT findings raise concern for venous sinus thrombosis or venous infarction.
- ☐ CTV acquisition and contrast administration follow the approved Radiology venography technique and institutional contrast-safety policy.
- ☐ CTV interpretation should document the involved dural venous sinus/cortical vein, extent of thrombosis, and associated venous infarction, edema, or hemorrhage when present.
- ☐ Critical CTV findings are communicated immediately to the Stroke/Neurology team and documented with the time of communication.
- ☐ When CTV is non-diagnostic or MRI is preferred/available, MRI with MR venography (MRV) may be used according to the institutional MRI Stroke Protocol.

6. CT Perfusion (CTP, when available/indicated)

6.1 Purpose

- ☐ Estimate ischemic core (irreversibly infarcted tissue) and penumbra (at-risk but potentially salvageable tissue) volumes
- ☐ Support core/penumbra mismatch-based patient selection for thrombectomy in the extended time window (commonly 6–24h from LKW) per DAWN/DEFUSE 3-based criteria
- ☐ Not required for early-window (0–6h) candidates who meet standard non-contrast CT/CTA criteria, per institutional selection protocol

6.2 Acquisition

- ☐ Dynamic cine acquisition through the region of interest during IV contrast bolus, per departmental CTP protocol and scanner-specific technique
- ☐ Post-processed using validated automated software (e.g., RAPID, Viz.ai, or institutional equivalent) to generate CBF, CBV, MTT/Tmax maps
- ☐ Automated core and penumbra volumes, and mismatch ratio, generated and transmitted to stroke team per departmental workflow

6.3 Key Perfusion Parameters

Map	Represents
Cerebral Blood Flow (CBF)	Rate of blood flow through tissue; severely reduced rCBF used to estimate irreversible core
Cerebral Blood Volume (CBV)	Total blood volume in tissue; reduced CBV correlates with infarct core
Mean Transit Time (MTT) / Tmax	Delay in contrast arrival/washout; prolonged Tmax identifies hypoperfused (core + penumbra) tissue
Core volume	Estimated irreversibly infarcted tissue (severely reduced CBF/CBV)
Mismatch volume / ratio	Penumbra (Tmax-defined hypoperfusion) minus core; used against DAWN/DEFUSE 3 thresholds for extended-window selection

Refer to institutional Mechanical Thrombectomy Selection Protocol for the specific core-volume and mismatch thresholds applied at each time window.

7. Automated LVO Detection & Notification Software

- ☐ Automated large-vessel-occlusion detection software (if available) configured to send direct alert/notification to stroke team, interventionalist, and radiologist simultaneously
- ☐ Automated software output is adjunctive only — final interpretation and clinical decision rest with the radiologist and stroke/neurointervention team, not the software flag alone
- ☐ Downtime/failure of automated notification does not replace standard radiologist read and direct verbal communication of critical results

8. Reading, Communication & Documentation

- ☐ Non-contrast CT interpreted and verbally communicated to stroke team STAT — do not wait for final formal report before verbal communication of hemorrhage status
- ☐ CTA/CTP findings communicated verbally to stroke/neurointervention team as soon as available, given time-critical treatment decisions
- ☐ Critical results (hemorrhage, LVO, large completed infarct) documented with time of identification and time of Communication

- [ ] Final formal radiology report completed per departmental turnaround policy, referencing ASPECTS, occlusion site, collateral status, and (if performed) core/penumbra volumes
- [ ] All relevant imaging times (order, acquisition start, images available, verbal communication, final read) documented for stroke registry/Get With The Guidelines–Stroke reporting

9. Quality Metrics to Track

- [ ] Door-to-CT (needle) initiation time
- [ ] Door-to-CT interpretation/verbal communication time
- [ ] Door-to-CTA completion time (for LVO detection)
- [ ] Imaging-to-groin-puncture time for thrombectomy patients
- [ ] Rate of contrast-related adverse events

Radiology / Stroke Team Sign-Off

Radiologist Signature: ____________________________ Date/Time: __________________

CT Technologist: ____________________________ Stroke Team Notified (name/time): _______________

References: 2026 AHA/ASA Guideline for the Early Management of Patients With Acute Ischemic Stroke and subsequent focused updates; AHA/ASA Target: Stroke initiative imaging time benchmarks; AHA/ASA Get With The Guidelines–Stroke measures; DAWN trial (Nogueira et al., NEJM 2018) and DEFUSE 3 trial (Albers et al., NEJM 2018) perfusion-based selection criteria; American Stroke Association International Stroke Center Certification program Primary/Comprehensive Stroke Center imaging turnaround requirements. This template must be re-verified against current departmental scanner protocols, contrast policies, and the current guideline version at the time of institutional adoption.